

MAX MUZEAU

Data Scientist · ML Engineer

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Data scientist and ML engineer specialized in **computer vision** and **deep learning** for **earth observation**. PhD from Onera and CentraleSupélec on anomaly detection in SAR images. Now at Geowatch Labs, working at the intersection of research, ML engineering and data engineering.

EXPERIENCE

Data Scientist · Geowatch Labs

Oct 2024 – Present

- Design and adapt SOTA **deep learning** and **computer vision** models for **satellite image analysis**.
- Drive the full **R&D-to-deployment cycle**: from prototyping algorithms to serving them at **global scale**, processing agricultural fields worldwide.
- Build scalable **geospatial data pipelines** on **Google Cloud** (Apache Beam, Dataflow, Airflow), containerized with Docker and orchestrated on Kubernetes.
- Preprocess, analyze and visualize **geospatial data** (QGIS, GeoPandas).

PhD Student · ONERA – CentraleSupélec

Oct 2021 – Oct 2024

- **Anomaly detection** in remote sensing data using **self-supervised learning** for computer vision.
- **Contrastive learning** with ViTs (vision transformers) to build **geospatial foundation models**.
- **Synthetic Aperture Radar (SAR)** data processing and analysis.
- **Statistical methods** (a contrario, maximum likelihood estimation) and robust estimation.
- Author of multiple **scientific publications** (see page 2).

Computer Science Teacher · IUT Orsay

Jan 2022 – Nov 2023

- Taught Python programming and algorithmics to undergraduate students.

Research Intern in SAR Imagery · Télécom Paris

Mar 2021 – Aug 2021

- SAR image denoising with deep learning and signal processing; activity monitoring of an island with Sentinel-1 images in collaboration with the CNES.

Artificial Intelligence Consultant · Lanteas

Aug 2020 – Feb 2021

- Built a deep learning algorithm to predict the profit of municipalities.

EDUCATION

PhD, Anomaly Detection Schemes for SAR Images · Université Paris-Saclay

2021 – 2024

Master OIVM – AIMA, advanced imaging (SAR, hyperspectral, LIDAR) · Jean Monnet University

2020 – 2021

Engineering Degree, computer vision & signal processing · Télécom Saint-Étienne

2018 – 2021

CPGE, intensive undergraduate course in mathematics & physics · Lycée Jacques Amyot

2016 – 2018

SKILLS

Languages Python, C++, MATLAB

ML / Deep learning PyTorch, TensorFlow, scikit-learn

Cloud & data pipelines Google Cloud, Dataflow, Apache Beam, Airflow

MLOps & deployment Docker, Kubernetes, GitHub Actions (CI/CD)

Geospatial QGIS, GeoPandas, shapely, Fiona, GDAL / rasterio

Data SQL, BigQuery, pandas

Dev tooling Poetry, Git, Linux, Bash

Spoken languages French (native), English (fluent)

SAFE: A SAR Feature Extractor Based on Self-Supervised Learning and Masked Siamese ViTs

M. Muzeau, J. Frontera-Pons, C. Ren, J.-P. Ovarlez — IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing

Anomaly Detection Schemes for SAR Images

M. Muzeau — PhD thesis, Université Paris-Saclay

General Feature Extraction in SAR Target Classification: A Contrastive Learning Approach Across Sensor Types

M. Muzeau, J. Frontera-Pons, C. Ren, J.-P. Ovarlez — IGARSS 2024, IEEE International Geoscience and Remote Sensing Symposium

Deep-Learning Based Wall Mitigation Method for Through-the-Wall Radar Imaging

Y. Boudrouz, M. Muzeau — 2024 International Radar Conference (RADAR)

On the Impact of Despeckling for Supervised SAR Super-Resolution

M. Muzeau, C. Ren, J. Fix, F. Brigui, J.-P. Ovarlez — EUSAR 2024, 15th European Conference on Synthetic Aperture Radar

Self-Supervised SAR Anomaly Detection Guided with RX Detector

M. Muzeau, C. Ren, S. Angelliaume, M. Datcu, J.-P. Ovarlez — IGARSS 2023, IEEE International Geoscience and Remote Sensing Symposium

Apprentissage auto-supervisé guidé statistiquement pour la détection d'anomalies en imagerie SAR

M. Muzeau, C. Ren, D. Mihai, J.-P. Ovarlez — GRETSI 2023

Self-Supervised Learning Based Anomaly Detection in Synthetic Aperture Radar Imaging

M. Muzeau, C. Ren, S. Angelliaume, M. Datcu, J.-P. Ovarlez — IEEE Open Journal of Signal Processing, vol. 3, pp. 440–449

SAR Anomalies Detection Based on Deep Learning

M. Muzeau, C. Ren, S. Angelliaume, M. Datcu, J.-P. Ovarlez — XXVIIIème Colloque GRETSI 2022

Self-Supervised Training Strategies for SAR Image Despeckling with Deep Neural Networks

E. Dalsasso, L. Denis, M. Muzeau, F. Tupin — EUSAR 2022, 14th European Conference on Synthetic Aperture Radar